

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	12 July 2026
Team ID	
Project Name	Customer Registry
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

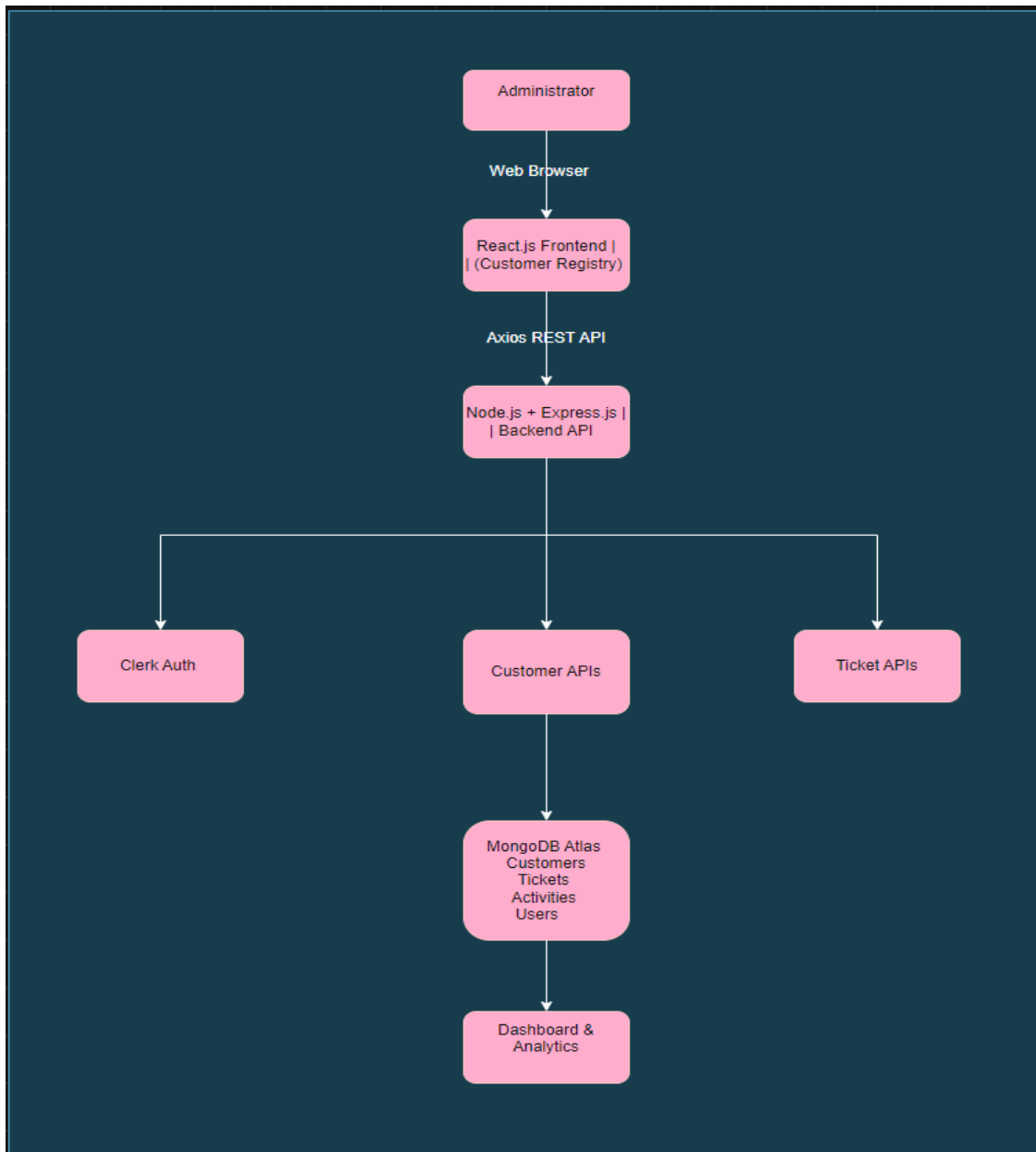


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Responsive web interface for administrators and customer care executives	React.js, HTML5, CSS3, JavaScript
2	Application Logic-1	User Authentication & Authorization	Node.js, Express.js, Clerk Authentication
3	Application Logic-2	Customer Management (Add, Edit, Delete, Search Customers)	Express.js, REST API
4	Application Logic-3	Ticket Management, Dashboard & Analytics	Express.js, Chart.js
5	Database	Stores users, customers, tickets and activity records	MongoDB (MongoDB Atlas)
6	Cloud Database	Cloud-hosted NoSQL database	MongoDB Atlas
7	File Storage	Stores uploaded customer documents (if any)	Local Storage / Cloud Storage
8	External API-1	User Authentication Service	Clerk API
9	External API-2	REST API Communication	Axios
10	Machine Learning Model	Not Applicable	N/A
11	Infrastructure (Server/Cloud)	Frontend & Backend Deployment	Render, GitHub

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used to build the application	React.js, Node.js, Express.js
2	Security Implementations	Secure authentication, password protection, API security and CORS	Clerk Authentication, HTTPS, JWT, CORS
3	Scalable Architecture	Three-tier architecture supporting future expansion	React + Express + MongoDB
4	Availability	Cloud-hosted application accessible anytime	Render, MongoDB Atlas
5	Performance	Fast API responses, optimized database queries, efficient dashboard rendering	Axios, MongoDB Indexing, React Vite